

Lecture 3: Covariance Estimation

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1 Covariance Estimation: Guarantees

Consider a mean-zero random vector $\mathbf{x} \in \mathbb{R}^d$. Its population covariance matrix is defined as

$$\Sigma = \mathbb{E}[\mathbf{x}\mathbf{x}^\top],$$

while the empirical covariance matrix based on n independent copies $\mathbf{x}_1, \dots, \mathbf{x}_n$ is

$$\Sigma_n = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top.$$

A central question is whether Σ_n provides a good approximation of Σ in operator norm, that is, whether

$$\Sigma_n \approx \Sigma.$$

The operator norm can be expressed as

$$\|\Sigma_n - \Sigma\| = \max_{\mathbf{v} \in S^{d-1}} |\mathbf{v}^\top (\Sigma_n - \Sigma) \mathbf{v}|,$$

where S^{d-1} is the set of vectors on the unit sphere.

To control this supremum, one introduces an ε -net $\mathcal{N} \subset S^{d-1}$, which may be chosen (see Vershynin (2018)) so that

$$|\mathcal{N}| \leq \left(\frac{2}{\varepsilon} + 1\right)^d = 9^d \quad \text{for } \varepsilon = \frac{1}{4}.$$

A standard covering argument Vershynin (2018) then implies that

$$\|\Sigma_n - \Sigma\| \leq \frac{1}{1 - 2\varepsilon} \max_{\mathbf{v} \in \mathcal{N}} |\mathbf{v}^\top (\Sigma_n - \Sigma) \mathbf{v}|.$$

With $\varepsilon = \frac{1}{4}$, this becomes

$$\|\Sigma_n - \Sigma\| \leq 2 \max_{\mathbf{v} \in \mathcal{N}} \left| \frac{1}{n} \sum_{i=1}^n \langle \mathbf{x}_i, \mathbf{v} \rangle^2 - \mathbb{E}[\langle \mathbf{x}, \mathbf{v} \rangle^2] \right|,$$

and, for convenience, we denote

$$S(\mathbf{v}) = \frac{1}{n} \sum_{i=1}^n \langle \mathbf{x}_i, \mathbf{v} \rangle^2.$$

This quantity represents the empirical variance of the one-dimensional projections $\langle \mathbf{x}_i, \mathbf{v} \rangle$ and plays a key role in bounding $\|\Sigma_n - \Sigma\|$. For each $\mathbf{v}$, $S(\mathbf{v})$ is a sum of independent random variables. We will therefore make use of Bernstein's inequality, stated below, to control it.

Theorem 1 (Bernstein's inequality). *If X_i are independent subexponential random variables:*

$$\Pr\left(\left|\sum_{i=1}^n (X_i - \mathbb{E}X_i)\right| > t\right) \leq 2 \exp\left[-c \min\left(\frac{t^2}{\sigma^2}, \frac{t}{K}\right)\right],$$

where

$$\sigma^2 = \sum_{i=1}^n \|X_i\|_{\psi_1}^2, \quad K = \max_i \|X_i\|_{\psi_1}.$$

Recall that $\|X\|_{\psi_1}$ is the subexponential norm defined as:

$$\|X\|_{\psi_1} := \inf \left\{ t > 0 : \mathbb{E} \left[\exp \left(\frac{|X|}{t} \right) \right] \leq 2 \right\}.$$

If all X_i are identically distributed, then $\sigma^2 = K^2 n$. Let $t = \delta n$. Then Bernstein's inequality further simplifies to:

$$\Pr \left(\left| \frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E} X \right| \geq \delta \right) \leq 2 \exp \left[-c \min \left(\frac{\delta^2 n^2}{K^2 n}, \frac{\delta n}{K} \right) \right] = 2 \exp \left[-c \min \left(\frac{\delta^2}{K^2}, \frac{\delta}{K} \right) n \right]. \quad (**)$$

We are now ready to state and prove the main theorem.

Theorem 2 (Covariance Estimation). *If $\mathbf{x}_i \sim \mathcal{N}(0, \Sigma)$ and $n > Cd$, then*

$$\|\Sigma_n - \Sigma\| \leq 0.1 \|\Sigma\|$$

with probability at least $1 - 2e^{-cn}$.

Proof. W.l.o.g. we assume that $\|\Sigma\| = 1$.

Fix any $\mathbf{v} \in S^{d-1}$ (unit sphere of $\mathbb{R}^d$).

Note that if $\mathbf{x}$ is a mean-0 normal random vector, then $\langle \mathbf{x}, \mathbf{v} \rangle$ is also a mean-0 normal random variable. Therefore:

$$\langle \mathbf{x}, \mathbf{v} \rangle \sim \mathcal{N}(0, \sigma^2).$$

Let us compute σ^2 :

$$\sigma^2 = \mathbb{E} \langle \mathbf{x}, \mathbf{v} \rangle^2 = \mathbf{v}^\top (\mathbb{E} \mathbf{x} \mathbf{x}^\top) \mathbf{v} = \mathbf{v}^\top \Sigma \mathbf{v} \leq \|\Sigma\| = 1.$$

The Gaussian tail property gives

$$\Pr \{ |\langle \mathbf{x}, \mathbf{v} \rangle| \geq t \} \leq 2 \exp \left(-\frac{t^2}{\sigma^2} \right) \leq 2 \exp(-t^2).$$

Then

$$\Pr \{ \langle \mathbf{x}, \mathbf{v} \rangle^2 \geq s \} = \Pr \{ |\langle \mathbf{x}, \mathbf{v} \rangle| \geq \sqrt{s} \} \leq 2 \exp(-s).$$

Thus $\langle \mathbf{x}, \mathbf{v} \rangle^2$ is subexponential and

$$\|\langle \mathbf{x}, \mathbf{v} \rangle^2\|_{\psi_1} \leq C.$$

Apply Bernstein to the random variables $X_i = \langle \mathbf{x}_i, \mathbf{v} \rangle^2$, with $K = C$, $\delta = 0.01$ yields

$$\Pr \left(\left| \frac{1}{n} \sum_{i=1}^n \langle \mathbf{x}_i, \mathbf{v} \rangle^2 - \mathbb{E} \langle \mathbf{x}, \mathbf{v} \rangle^2 \right| \geq 0.01 \right) \leq 2 \exp(-c'n).$$

Let

$$S(\mathbf{v}) := \frac{1}{n} \sum_{i=1}^n \langle \mathbf{x}_i, \mathbf{v} \rangle^2 - \mathbb{E} \langle \mathbf{x}, \mathbf{v} \rangle^2.$$

Using the ε -net $\mathcal{N}$ with $|\mathcal{N}| \leq 9^d$, we obtain:

$$\Pr \left\{ \max_{\mathbf{v} \in \mathcal{N}} |S(\mathbf{v})| \geq 0.01 \right\} \leq |\mathcal{N}| \cdot 2 \exp(-c'n) \leq 2 \exp \left(-\frac{c'}{2} n \right) \quad \text{if } d \leq c''n,$$

where the first inequality is a simple application of the union bound.

The complement holds with probability $\geq 1 - 2 \exp(-c'n/2)$. When it holds:

$$\|\Sigma_n - \Sigma\| \leq 2 \max_{\mathbf{v} \in \mathcal{N}} |S(\mathbf{v})| \leq 2 \cdot 0.01 = 0.1.$$

□

2 Perturbation Theory

Eigenvalues So far, we have established that Σ_n provides a good approximation of Σ . We now show that this operator-norm control implies corresponding guarantees on the spectral structure, namely

$$\lambda_i(\Sigma_n) \approx \lambda_i(\Sigma), \quad \mathbf{v}_i(\Sigma_n) \approx \mathbf{v}_i(\Sigma),$$

where $\lambda_i(\mathbf{A})$ and $\mathbf{v}_i$ is the i -th eigenvalue and corresponding eigenvector of the matrix $\mathbf{A}$.

Theorem 3 (Weyl's inequality). *For all $d \times d$ symmetric matrices $\mathbf{A}, \mathbf{B}$,*

$$\max_i |\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})| \leq \|\mathbf{A} - \mathbf{B}\|.$$

Proof sketch. The key idea is to use the variational (Courant–Fischer) characterization of eigenvalues:

$$\lambda_i(\mathbf{A}) = \min_{\dim U=i} \max_{\substack{\mathbf{x} \in U \\ \|\mathbf{x}\|=1}} \mathbf{x}^\top \mathbf{A} \mathbf{x}. \quad (\text{CF})$$

Applying the same formula to $\mathbf{B}$ and comparing the two expressions shows that replacing $\mathbf{A}$ by $\mathbf{B} = \mathbf{A} + (\mathbf{B} - \mathbf{A})$ perturbs every Rayleigh quotient by at most $\|\mathbf{A} - \mathbf{B}\|$.

Define $\mathbf{E} = \mathbf{B} - \mathbf{A}$. Let $U_{\mathbf{A}}$ be a subspace of dimension i that attains the minimum in (CF) for $\mathbf{A}$. Then

$$\lambda_i(\mathbf{A}) = \max_{\substack{\mathbf{x} \in U_{\mathbf{A}} \\ \|\mathbf{x}\|=1}} \mathbf{x}^\top \mathbf{A} \mathbf{x}. \quad (1)$$

Since the minimization in (CF) is over all dimension- i subspaces, we have

$$\lambda_i(\mathbf{B}) \leq \max_{\substack{\mathbf{x} \in U_{\mathbf{A}} \\ \|\mathbf{x}\|=1}} \mathbf{x}^\top \mathbf{B} \mathbf{x}. \quad (2)$$

Substituting $\mathbf{B} = \mathbf{A} + \mathbf{E}$ gives

$$\mathbf{x}^\top \mathbf{B} \mathbf{x} = \mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{x}^\top \mathbf{E} \mathbf{x}.$$

Applying this inside the maximum in (2) yields

$$\lambda_i(\mathbf{B}) \leq \max_{\substack{\mathbf{x} \in U_{\mathbf{A}} \\ \|\mathbf{x}\|=1}} \mathbf{x}^\top \mathbf{A} \mathbf{x} + \max_{\|\mathbf{x}\|=1} \mathbf{x}^\top \mathbf{E} \mathbf{x}.$$

By (1) and the fact that the second term equals the operator norm $\|\mathbf{E}\|$,

$$\lambda_i(\mathbf{B}) \leq \lambda_i(\mathbf{A}) + \|\mathbf{E}\|.$$

This proves

$$\lambda_i(\mathbf{B}) - \lambda_i(\mathbf{A}) \leq \|\mathbf{A} - \mathbf{B}\|.$$

Exchanging the roles of $\mathbf{A}$ and $\mathbf{B}$ gives the reverse inequality

$$\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B}) \leq \|\mathbf{A} - \mathbf{B}\|.$$

Combining these two bounds yields

$$|\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})| \leq \|\mathbf{A} - \mathbf{B}\|.$$

□

Eigenvectors Eigenvectors can only be recovered up to a global sign, and their stability under perturbations fundamentally depends on the presence of a *spectral gap*. When the gap $\lambda_k(\mathbf{A}) - \lambda_{k+1}(\mathbf{A})$ is small, even a tiny perturbation of the matrix can cause the associated eigenvectors to rotate significantly or swap order. The Davis–Kahan theorem formalizes this dependence by controlling the distance between the subspaces spanned by the top k eigenvectors of two symmetric matrices in terms of both the perturbation size and the spectral gap.

Theorem 4 (Davis–Kahan inequality). *Let $\mathbf{A}, \mathbf{B}$ be $d \times d$ symmetric matrices, and $1 \leq k \leq d$. Let*

$$P_{\mathbf{A}} = \text{orthogonal projection onto } \text{span}\{\mathbf{v}_1(\mathbf{A}), \dots, \mathbf{v}_k(\mathbf{A})\},$$

$$P_{\mathbf{B}} = \text{orthogonal projection onto } \text{span}\{\mathbf{v}_1(\mathbf{B}), \dots, \mathbf{v}_k(\mathbf{B})\}.$$

Then

$$\|P_{\mathbf{A}} - P_{\mathbf{B}}\| \leq \frac{\|\mathbf{A} - \mathbf{B}\|}{\lambda_k(\mathbf{A}) - \lambda_{k+1}(\mathbf{A})}.$$

Proof sketch. The proof is based on decomposing the perturbation $\mathbf{B} - \mathbf{A}$ in the eigenbasis of $\mathbf{A}$ and examining how it mixes the top- k eigenspace with its orthogonal complement. A resolvent identity shows that the "rotation" of the invariant subspace is proportional to the off-diagonal blocks of the perturbation, while the spectral gap controls how strongly these blocks influence the projection operators. $\square$

Combining the covariance-estimation theorem with Weyl's inequality and the Davis-Kahan bound shows that sample PCA is close to population PCA whenever the sample size n is at least on the order of the dimension d , and the population covariance $\mathbf{\Sigma}$ possesses a nontrivial eigengap separating its top k eigenvalues from the remainder of the spectrum.

References

Roman Vershynin. *High-dimensional probability: An introduction with applications in data science*, volume 47. Cambridge university press, 2018.